

MGTF 430: ASSET MANAGEMENT

Spring 2026, Master of Quantitative Finance

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DESCRIPTION

This course will provide a rigorous exploration of the use of quantitative tools in asset management, bridging theory with real world investment practice in the quest for generating superior risk-adjusted returns. The primary focus will be equities. The course will have a considerable programming requirement and will utilize Python.

OBJECTIVES

At the close of this course, you will be able to:

- Utilize a comprehensive framework for quantitative investment research.
- Exhibit practical skills in equity investment signal generation.
- Implement quantitative investment research tasks in Python.
- Navigate practical issues a portfolio manager faces in implementing a strategy.
- Rigorously evaluate investment performance of a strategy and put it in context.
- Present investment ideas and results in a clear, concise, and organized framework.

MATERIALS

Required:

- Chincarini, L., & Kim, D. (2023). *Quantitative Equity Portfolio Management* (2nd ed.). McGraw-Hill.
- Grinold, R. C., & Kahn, R. N. (2000). *Active Portfolio Management* (2nd ed.). McGraw-Hill.

REQUIRED SOFTWARE

- Each student will need a computer for the Python-based homework and final project.
- These computers should be running Python 3.8 or higher and Jupyter notebook.
- Students can use the Anaconda distribution for Python and Anaconda Navigator, VS Code, etc.
- Google Sheets may be occasionally used for calculation verification purposes.
- LLM tools (such as ChatGPT or equivalent) may be occasionally used.

SCHEDULE

Week	Date	Class Topic & Activities	Due
1	April 3 rd	Introduction and Investing Landscape	
2	April 10 th	Theoretical Foundations	
3	April 17 th	Signal Generation: From Insight to Alpha	Homework #1
4	April 24 th	Signal Research	Homework #2
5	May 1 st	Midterm Exam; Signal Research II	Final Project Proposal
6	May 8 th	Portfolio Construction	Homework #3
7	May 15 th	Performance Measurement & Attribution	
8	May 22 nd	AI in Investing; Contemporary Topics	Homework #4
9	May 29 th	Active ETFs; International Investing	Final Project
10	June 6 th	Final Project Presentations; Course Review	Homework #5

ASSIGNMENTS

There will be 5 assignments. Collaboration with classmates is allowed. However, each student must submit their own work.

CLASS ATTENDANCE

In-person class attendance is mandatory. Participating in class, such as asking questions and speaking up in class discussions are highly encouraged, however it is not mandatory for receiving a full 'class attendance' grade. After the first class, there will be a simple question to answer at the end (with all electronic devices put away) that will be easy to answer for those paying attention to the class. Answering the question correctly will be needed for full credit for class attendance.

MIDTERM EXAM

There will be a midterm held during the 5th week. It will be closed-book, closed-notes, and in-person. Computer usage will not be allowed for the exam.

FINAL PROJECT

Groups will be determined during the first two weeks of the course. Project subject proposals will be due on the 5th week of the course. Students are expected to propose, test, and evaluate an original quantitatively-based investment signal. Groups will need to submit their project report and python code by the due date, and will present their project to the class the following week.

GRADING

Category	Percentage
Class Attendance	10
Homework	25
Midterm Exam	25
Final Project	40

CHAT-GPT AND RELATED TOOLS

Students may use large language model (LLM) tools (e.g., ChatGPT, Gemini, Claude, Grok, Perplexity, etc.) to support their learning, homework assignments, and project work.

However, all submitted work must reflect the student's own understanding, analysis, and effort. Reliance on LLM-generated content in bulk or without critical engagement is not permitted. Moreover, certain assignments may restrict or prohibit the use of LLMs, which will be clearly noted in the assignment instructions when applicable.

Students are responsible for verifying the accuracy and appropriateness of any AI-generated material and must be able to explain and justify all aspects of their work. This policy aligns with professional expectations in the workplace, where AI tools typically may be used to assist with tasks, but individual accountability, accuracy, and judgment remain essential.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.